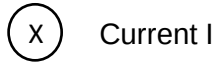
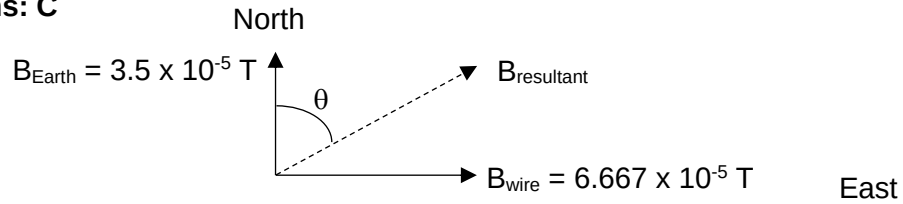


21. Ans: C



By right hand grip rule, magnetic field produced by wire at compass, B_{wire} , points to east.

$$\begin{aligned} B_{\text{wire}} &= \frac{\mu_0 I}{2\pi r} \\ &= \frac{4\pi \times 10^{-7} \times 3.0}{2\pi \times 9 \times 10^{-3}} \\ &= 6.667 \times 10^{-5} \text{ T} \end{aligned}$$

$$\tan\theta = \frac{B_{\text{wire}}}{B_{\text{Earth}}}$$

$$\begin{aligned} \theta &= \tan^{-1} \frac{6.667}{3.5} \\ &= 62^\circ \end{aligned}$$

